

A Swarm Intelligent Framework for Knowledge Transfer Between Heterogeneous Machine Learning Models in Multi Robot Systems

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Abstract

The paradigm of swarm intelligence and multi-agent systems has proved to be useful to obtain scalable, decentralized and collaborative decision making in distributed systems. Heterogeneous machine learning models used in multi-agent systems, however, widely operate independently which results not only in redundant inferences but also in higher computational load with a lack of reuse of knowledge. The solutions presently available are mostly based on homogenous architectures and collaboration at the training level, they aren't fit for real-time knowledge sharing between heterogeneous models.

This work introduces a model-agnostic, swarm-intelligent approach for knowledge transfer between various ML models. This proposed system utilizes three heterogeneous perception agents, all of which are trained on human detection from video streams, using either the MobileNet or the EfficientNet or the ResNet architecture. Each agent produces "knowledge packets" that are structured and carry semantic data like object labels, confidence scores, feature embeddings, and more, including agent identifiers, and timestamps. The packets are distributed by a collective memory module where a distributed communication mechanism is used.

Agents are used to store and retrieve knowledge packets from the collective memory for every video frame so that knowledge can be re-used and avoid redundant processing. The framework also includes an adaptation mechanism where agents can learn from shared knowledge representations, enhancing the agents' perception process over time. The evaluations while experimenting have shown that the knowledge transfer between heterogeneous models leads to improved collaborative perception, less redundant inference, efficient knowledge reuse and better performance. The proposed framework establishes the feasibility of real-time knowledge sharing among multi-agent systems with heterogeneous architectures and this brings a base for future swarm-intelligent robotic systems, such as autonomous navigation, surveillance, distributed sensing, and collaborative AI systems.

Keywords

Swarm Intelligence, Knowledge Transfer, Multi-Robot Systems, Collective Memory, Heterogeneous Machine Learning Models, Collaborative Intelligence.

1. Introduction

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and autonomous robotic systems has significantly increased the adoption of multi-agent and multirobot environments across various application domains. The modern systems that's robotic are being deployed in surveillance, autonomous navigation, disaster response, industrial automation, and various applications also including smart city, where multiple agents must collaborate to achieve common objectives efficiently [2], [6]. To enable such collaboration, researchers have drawn inspiration from swarm intelligence, a computational paradigm based on the cumulative behaviour of many social organisms such as ants, bees, and bird flocks. The distinguishing features of swarm-intelligent systems are decentralization, self-organization, scalability, and fault-tolerance, that makes them ideal for dynamic and distributed environments [2, 9]. Thanks to the recent advancement in the field of machine learning and deep learning, the autonomous agents are able to perform advanced perception processes such as object detection, image classification, scene understanding and the decision making with a very good accuracy [5], [11], [12]. Yet, in many cases, the choice of machine learning architecture used by different agents varies because of variance in hardware resources, energy consumption, and deployment requirements. Hence, nowadays, multi-agent systems tend to be more heterogeneous, consisting of agents with different computational power and perception models [6, 7, 8].

While heterogeneous models are flexible and fit for efficient use of resources, they also come with a number of difficulties to collaborate and share knowledge. Most current systems involve separate perception by agents even in similar situations, and even in the case of identical objects. This results in duplicate calculations, slower inference time, larger resource use and ineffective use of the collective intelligence present in the swarm [3, 5]. Moreover, lightweight models which run on resource-constrained agents cannot directly take advantage of the knowledge produced by more powerful or

accurate models [1, 6]. To enable the collaboration of several agents without compromising data privacy, a number of distributed learning varieties have been proposed, such as federated learning and swarm learning [1] and [4]. These strategies, however, are mainly aimed at collaborative model training and tend to make simplifying assumptions about having homogeneous model architectures. Likewise, knowledge distillation methods facilitate the transfer of knowledge from the big model to the small model, but they can be mostly used for retraining and centralized processing and are not appropriate for real-time collaborative environments [6], [10].

The limitations expose the necessity of a decentralized and model agnostic framework to share useful perception knowledge between heterogeneous machine learning models for inference. This can make a effective use of resources, less redundant processing and emergence of collective intelligence on the swarm level, without the need of raw data exchange or sharing of model parameters, respectively [1, 2, 10]. The work proposed in this project aims to solve this problem by developing a swarm-intelligent approach for transferring structured knowledge packets from one or more autonomous agents to the next and enhancing the perception of the autonomous agents through a collective memory mechanism.

The major importance of this research is summarized as follows:

1. A swarm-intelligent framework for a decentralized network for knowledge transfer among heterogeneous machine learning models.
2. A structured knowledge packet mechanism for efficient inter-agent communication.
3. A collective memory architecture that stores and distributes high-confidence perception knowledge.
4. A model-agnostic knowledge-sharing strategy that eliminates the need for raw data or parameter exchange.
5. An experimental evaluation demonstrating the effectiveness of collaborative knowledge transfer in improving decision consistency and knowledge reuse.

2. Literature Review

Swarm intelligence, distributed and federated AI and learning, collaborative AI and optimization, and decentralized optimization have a major role in advancing the development of intelligent multi-agent systems. The whole idea of mechanisms enabling multiple autonomous agents to cooperate, exchange information and solve jointly complex tasks in a scalable and adaptive manner has started to gain more importance in the research community. While significant advances are made, but there is a problem of transferring knowledge between heterogeneous machine learning models effectively. This section summarizes the most relevant work in the field of swarm Intelligence, distributed learning, heterogeneous systems and collaborative knowledge sharing.

Yang et al. [1] designed an efficient UAV swarm based multi-task federated learning framework for dynamic aerial environment. The framework allows decentralized learning where multiple UAVs learn task-specific knowledge collectively, and maintains data privacy. Dynamic task allocation mechanism and task-affinity-based knowledge sharing strategy were introduced to enhance convergence speed and the efficiency of learning. The experimental results showed that the task execution was improved, the communication overhead was decreased, and the utilization of the swarm resources was increased. The study emphasizes the use of the swarm intelligence to facilitate cooperation between multiple autonomous agents, by integrating it with federated learning. The study focuses on the integration of swarm intelligence and federated learning to facilitate the cooperation between multiple autonomous agents. The framework, however, assumes that the learning architectures of the participating UAVs are homogeneous and heavily depends on federated training procedures. So, most of the time knowledge sharing is done in the process of model training and not in real-time inference. As a result, the framework is not very effective in enabling collaboration among different ML models using different architectures. This research delivers interesting insights in small group of decentralized coordination, communication efficient collaboration and swarm level learning. We now propose a framework on decentralized collaboration but extend it to model-agnostic inference-level knowledge transfer between heterogeneous agents, without any federated retraining.

Kazim et al. [2] gave a very thorough review of Multi-Agent Swarm Engineering Systems (MASSES), covering the key points of designing Multi Agents, deployment, scalability issues, and coordination mechanisms. The study recognized that the fundamental elements of successful swarm systems are decentralization, self organization, robustness, adaptability and fault tolerance. In addition, the authors explored the potential of incorporating AI methods to enhance swarm decision-making and adaptability in large-scale autonomous environments. The work provides a valuable reference point for the design of robust architectures for the swarm that will run under dynamic and uncertain conditions. It is still at a higher level, from the system architecture perspective, and lacks of concrete solutions for the transfer of the learned knowledge from different heterogeneous machine learning models. This study inspires our project, based on the idea of decentralised architecture and coordination principles. We extend these ideas by adding a semantic knowledge sharing layer, which relies on structured knowledge packets and collective memory, for efficient perception knowledge sharing among heterogeneous agents.

Saadouni et al. [3] carried out a review of the IDS employed along with bio-inspired optimization algorithms and ML techniques in IoT environments. The study compared various swarm-based techniques and came to a end conclusion that there is a good gain in the feature selection and classification accuracy, computational speed and scalability. Some of the techniques like Particle Swarm

Optimization, Ant Colony Optimization and Genetic Algorithms were found to be betterment in performance of machine learning models in distributed environments. The examples in the review demonstrate that swarm intelligence can be used to enhance the distributed decision-making and optimization processes. However, the majority of the surveyed methods are still domain-specific, and they don't consider much of collaborative sharing of knowledge by the agents, which are heterogeneous. The results highlight the value of swarm-inspired collaboration and distributed intelligence. We integrate these ideas with the concept of a swarm to support knowledge sharing in a generalized way across a set of heterogeneous perception models, going beyond the cyber security context.

Srinivasan et al. [4] presented a federated meta-heuristic cryptography framework to enhance secure cloud data migration. By combining fuzzy logic, machine learning, reinforcement learning, and swarm-inspired optimization techniques, the framework enhances data security, reliability, and efficiency, benefiting from the positives of each method. The framework uses the strengths of each approach to ensure data security, reliability, and efficiency, using fuzzy logic, machine learning, reinforcement learning, and swarm-inspired optimization techniques.

communication efficiency. The results post experiment showed that the system achieved better performance in terms of throughput, packet delivery ratio, and secure data transfer performance. The framework emphasizes the advantages of decentralized learning paradigms but the main focus is secure communication in cloud rather than collaborative perception and knowledge transfer. Also, it takes relatively homogenous learning structures and neglects heterogeneous machine learning environments. Our framework took inspiration from the decentralized philosophy presented in this work. But our aim is specially geared towards facilitating cross-model knowledge exchange without the necessity for shared training processes, cryptographic synchronization and parameter sharing.

Nizamudeen [5] proposed a DL based intrusion detection system (IDS) for multi-cloud IoT design using swarm. The system proposed integrates swarm optimization approach and deep neural networks to enhance the classification accuracy and adaptability in distributed system. The framework achieved high detection accuracy or we can say performance while keeping low false-positive rates. This study demonstrates the importance or like how effective the swarm intelligence is in enhancing deep learning systems. However, the framework is designed for a specific security application and employs homogeneous learning ML models which are optimized for a single task. It does not investigate knowledge reuse or collaboration among heterogeneous machine learning architectures. The work contributes valuable insights regarding swarm-assisted intelligence enhancement. Our framework will generalize such concepts and applies them to task independent, model-agnostic knowledge transfer among multiple heterogeneous agents.

He et al. [6] presented a survey of multimodal large-model-driven perception systems for UAV swarms operating in complex low-altitude environments. The study emphasizes adaptive edge intelligence, collaborative sensing, federated evolution, and swarm-level perception. The authors identified data heterogeneity, scalability, online adaptation, and collaborative intelligence as major research challenges in next-generation UAV systems. One of many significant observations from this survey is the growing heterogeneity of perception models in autonomous systems. However, collaboration is primarily discussed in terms of coordinated sensing and perception rather than explicit semantic knowledge exchange. This work strongly motivates our research by validating the need for collaboration among heterogeneous learning models. We fill the gap by adopting a structured mechanism to exchange perception knowledge between heterogeneous agents, called a knowledge packet.

Zhang et al. [7] introduced the Hybrid Particle Swarm Optimization and Dynamic Clustering based energy efficient routing algorithm for heterogeneous wireless sensor networks. The routing mechanism dynamically adjusts the route based on the network conditions to achieve better energy balance and prolong network life. While the study is not about machine learning, it highlights the benefits of adaptation in the cooperation of heterogeneous agents with limited resources in the context of communication efficiency. The ideas of resource-aware collaborations hold true in our project and are extended to the sharing of knowledge between machine learning agents.

A clustering algorithm based on genetic algorithm was proposed in [8] for resource allocation in a heterogeneous fog computing platform. The framework substantially reduced the clustering error, reduced latency, and enhanced all resource utilization on distributed fog infrastructures. The study points to the effective and efficient strategies for handling the heterogeneity by decentralized optimization. It doesn't, however, solve any knowledge transfer or collaborative learning among machine learning models semantically. We approach the design of our framework according to the heterogeneity-aware design philosophy introduced in this work, and extend it to the semantic knowledge exchange between heterogeneous perception agents.

Achouri et al. [9] integrated the advantages of SWA with neural networks for SHM systems. By bringing together optimization techniques with deep learning models, the proposed hybrid framework demonstrated better prediction accuracy and convergence properties. The study presents the complementarity of swarm intelligence with ML systems in terms of their robustness and performance. But it is still application oriented and application specific, which is not applicable in distributed collaborative environments. We extend this synergy and turn its use towards decentralized, real-time perception systems consisting of several autonomous agents.

Amico et al. [10] proposed a decentralized reinforcement learning approach where multiple language models can share learning experiences via the shared rollout. The proposed approach showed remarkable performance enhancements without the necessity of a centralized parameter synchronization. This paper is relevant especially in introducing the idea of sharing high level experience instead of raw data or model parameters. The results show that the sharing of semantic level knowledge can highly enhance learning efficiency. We follow the same principle in our framework and share structured knowledge packets instead of raw sensor data or model parameters with other heterogeneous perception models.

An intelligent waste classification framework was proposed by Sayed et al. [11] that brings together deep learning with multi-objective Beluga Whale Optimization. Experimental results showed classification accuracy and computational efficiency improvements. The framework shows that the effectiveness of swarm-assisted intelligence enhancement is valid but still has a restriction in the application domain and includes centralized learning mechanisms. The study shows the importance of using swarm intelligence to enhance learning systems and our framework extends the notion of swarm intelligence to decentralized collaborative knowledge sharing.

Kishore et al. [12] proposed a Particle Swarm Optimisation based BiLSTM model for IDS in IoV. The learning performance and attack detection accuracy were significantly enhanced by incorporating PSO. Even if effective, the framework cannot be applied to heterogeneous model collaboration and cross-agent knowledge reuse. What knowledge remains is limited to learning models. To the highest of our knowledge, this paper expands the previous swarm-enhanced learning ideas to include an agent-independent transfer mechanism that enables knowledge to be transmitted between heterogeneous machine learning agents.

Rao et al. [13] introduced an intelligent security system for IIOT environments based on ensemble machine learning models optimized using swarm intelligence. Experimental results showed better detection accuracy than the traditional ones, and the system was more robust. The proposed framework successfully integrates swarm intelligence and ensemble learning, but its emphasis is on optimizing the parameters of the application itself instead of transferring knowledge between different heterogeneous agents. The study also confirms the successful application of swarm based intelligence and inspires the architecture of cooperative knowledge sharing mechanisms that can be utilized to enable a variety of ML frameworks.

The literature reviewed shows significant advances in swarm intelligence, federated learning, distributed artificial intelligence, optimization and collaborative machine learning. Current methods work well to enhance the scalability, robustness, resource usage and learning outcomes. But there are some key issues that have yet to be

addressed. Existing frameworks typically make the assumption of homogeneous model architectures, rely on parameter synchronization, mandate collaborative retraining, or only optimize, not reuse knowledge.

Moreover, existing research is largely devoted to a single type of machine learning model. In addition, few existing studies focus on real-time collaboration between heterogeneous machine learning models at the inference level. There is significant research gap in the creation of a decentralized, model agnostic framework that allows heterogeneous machine learning models to share semantic information without the necessity to share raw data, training sets, and internal model parameters. To fill this gap is the main motivation for the proposed swarm-intelligent framework comprising of a structured knowledge packet-based communication mechanism, memory sharing, and collaborative knowledge re-use among heterogeneous agents.

2.1 Review Analysis and Research Gap

The literature asks suggests that there has been substantial advances in the field of Swarm intelligence, Federated learning, Distributed optimization and Collaborative machine learning systems. A number of existing works have shown how to enhance the scalability, adaptability, and system performance for multi-agent environments using decentralized decision making, swarm based coordination, and distributed learning [1]–[5] and [9]. But there are numerous key research gaps which haven't been answered yet.

The approaches in collaborative learning most commonly used, such as federated learning and swarm learning, mainly emphasize model training and mostly assume homogeneous machine learning architectures in the collaborating agents [1, 4]. These assumptions restrict their effective use in real-world multi-robot systems where the agents might make different ML models based on their computational power and hardware limitations [6] [8]. Further, current methods focus on parameter sharing, gradient aggregation, or centralized knowledge distillation [1, 4, 10] methods which are not applicable in real-time inference-level collaboration. As a result, agents keep on making the same inferences even though they might have generated these inferences in another part of the swarm, incurring unnecessary computational overhead, latency and resource usage [3, 5].

While many studies have been done on either resource allocation, routing, clustering or optimization in distributed systems, sensor networks and fog computing environments the main concern was the allocation of resources or routing, clustering or optimizing, rather than exchanging semantic knowledge between heterogeneous machine learning models [7, 8]. In the same way, recent papers in swarm engineering and collaborative perception focus on communication and coordination without offering a general swarm perception sharing mechanism between the various model architectures [2, 6]. Moreover, current approaches are not based on a standard and model independent representation that allows

heterogeneous agents to understand, save and apply knowledge produced by other agents. Without it, true collective intelligence does not develop: Agents would always be able to learn from the experiences and observations of the entire swarm [6], [10], [13].

Therefore, there is a necessity for a research gap to develop a decentralized infrastructure that facilitates knowledge/data transfer among heterogeneous machine learning models at the inference level in real time. This is the main motivation for the work proposed, which proposes the use of structured knowledges packets and a collective memory mechanism to allow an efficient knowledge sharing, the elimination of redundant inference, efficient use of the resources and the emergence of swarm level collective intelligence in multi-agent systems.

3. Materials and Methods

3.1 System Overview

The proposed framework introduces a swarm-intelligent knowledge transfer mechanism that enables heterogeneous machine learning models operating within a multi-robot environment to collaboratively share learned knowledge without exchanging raw data or model parameters. The framework consists of multiple autonomous agents equipped with different deep learning architectures, including MobileNet, ResNet, and EfficientNet. Each agent independently analyzes incoming visual data and generates structured knowledge packets that are stored within a shared collective memory. High-confidence knowledge is subsequently disseminated among participating agents to facilitate collaborative decision-making and reduce redundant inference operations.

The overall workflow consists of five major stages:

1. Data acquisition and preprocessing.
2. Independent inference by heterogeneous learning agents.
3. Knowledge packet generation.
4. Collective memory construction and knowledge selection.
5. Swarm-based knowledge dissemination and reuse.

This architecture enables decentralized collaboration while preserving the independence of individual learning models.

3.2 Proposed Swarm-Intelligent Knowledge Transfer Framework

The proposed architecture is a decentralized swarm-intelligent system that facilitates the transfer of knowledge between various heterogeneous ML models as autonomous agents. The proposed framework enables multiple ML models to interact through the sharing of high-level semantic knowledge during inference, in contrast to the existing centralized processing-based or homogeneous models or parameter-sharing approaches. The system has three heterogeneous perception agents which are MobileNet, EfficientNet and ResNet, one by one processing the video frames and generating the prediction results. The first sprint of the framework is the acquisition of video files and the extraction of the frames. All three agents take all the extracted frames and run local inference to determine the

presence of a human and compute the associated confidence scores. The output of the inferences are converted to knowledge packets using a standard format: frame ID, object label, confidence score, time stamp, and source agent information. These packets contain a common representation, regardless of the underlying model architecture used by all agents.

The knowledge packets produced are stored in a Collective Memory module, which becomes a shared knowledge repository for the swarm. In each frame, the packet that has the greatest level of confidence is chosen as the best representation of its knowledge and kept in memory. This best knowledge is then disseminated to other agents by a decentralized communication mechanism. Without the need for retraining, parameter sharing, or raw data sharing, receiving agents use the common knowledge to update their confidence and boost prediction quality.

The proposed system facilitates knowledge reuse between different and multiple models, minimizing redundant inferences, better resource utilization, increased prediction accuracy and collaborative intelligence. The framework is also decentralized, which enhances its scalability, fault-tolerance, and adaptability, thereby making it suitable for real-world multi-robot and distributed AI applications.

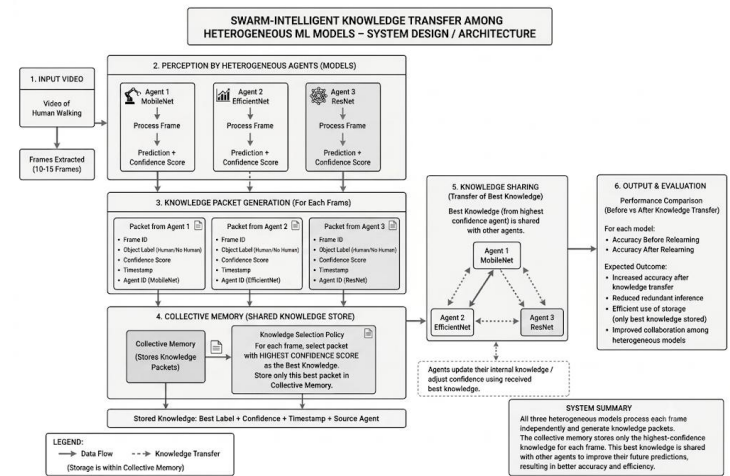


Figure 3.1: Proposed swarm-intelligent knowledge transfer architecture for collaborative learning among heterogeneous machine learning models. MobileNet, EfficientNet, and ResNet independently process video frames and generate knowledge packets containing prediction labels, confidence scores, timestamps, and agent identifiers. The generated packets are evaluated within a collective memory module, where the highest-confidence packet is selected and stored as the best knowledge. The selected knowledge is subsequently shared among participating agents to improve future predictions and reduce redundant inference operations.

To demonstrate a mathematical understanding of the proposed framework, the knowledge transfer process is shown using a common formulation for heterogeneous agents. This describes how individual predictions are converted into knowledge packets, tested based on confidence, and chosen for storage in the collective memory. The selected high confidence knowledge packet is then distributed among the agents, enabling collaborative knowledge reuse without exchanging raw data or model parameters.

The proposed framework enables heterogeneous perception agents to share the most reliable knowledge without exchanging raw data or model parameters. Consider a set of agents

$$\mathcal{A} = \{A_1, A_2, A_3\}, \quad (1)$$

where A_1 , A_2 , and A_3 correspond to MobileNet-V3 Small, EfficientNet-B0, and ResNet18, respectively. Let the input video be decomposed into T frames,

$$\mathcal{X} = \{X_1, X_2, \dots, X_T\}. \quad (2)$$

For any frame $X_t \in \mathcal{X}$, each agent $A_i \in \mathcal{A}$ independently performs inference using its own model and produces a prediction

$$P_{i,t} = f_i(X_t) = (l_{i,t}, c_{i,t}), \quad (3)$$

where $f_i(\cdot)$ is the inference function of agent A_i , $l_{i,t} \in \{\text{Human, No Human}\}$ is the predicted label, and $c_{i,t} \in [0, 1]$ is the confidence score associated with that prediction.

Since the agents are heterogeneous, their outputs are transformed into a common knowledge representation (knowledge packet) to enable comparison and sharing:

$$K_{i,t} = (t, l_{i,t}, c_{i,t}, a_i), \quad (4)$$

where t is the frame identifier and a_i denotes the source agent. For each frame, the collective memory receives the set of knowledge packets

$$\mathcal{K}_t = \{K_{1,t}, K_{2,t}, K_{3,t}\}. \quad (5)$$

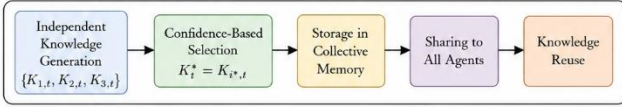
The collective memory selects the most reliable knowledge according to the highest confidence value:

$$i^* = \operatorname{argmax}_{A_i \in \mathcal{A}} c_{i,t}, \quad (6)$$

$$K_t^* = K_{i^*,t}, \quad (7)$$

$$c_t^* = \max_{A_i \in \mathcal{A}} c_{i,t}. \quad (8)$$

The selected packet K_t^* is stored in the collective memory and shared with all other agents. Thus, every agent can access the strongest perception knowledge available within the swarm. The knowledge-sharing process for each frame is summarized as



Mathematically, the core objective of the framework is expressed as

$$\max_{A_i \in \mathcal{A}} c_{i,t} \rightarrow c_t^* \rightarrow \text{shared swarm knowledge}, \quad (9)$$

which shows that multiple independent confidence estimates are reduced to a single collective confidence value c_t^* . Repeating this process for all frames $t = 1, 2, \dots, T$ enables the swarm of heterogeneous models to collaboratively reuse high-confidence knowledge, improving overall perception performance while reducing redundant computations.

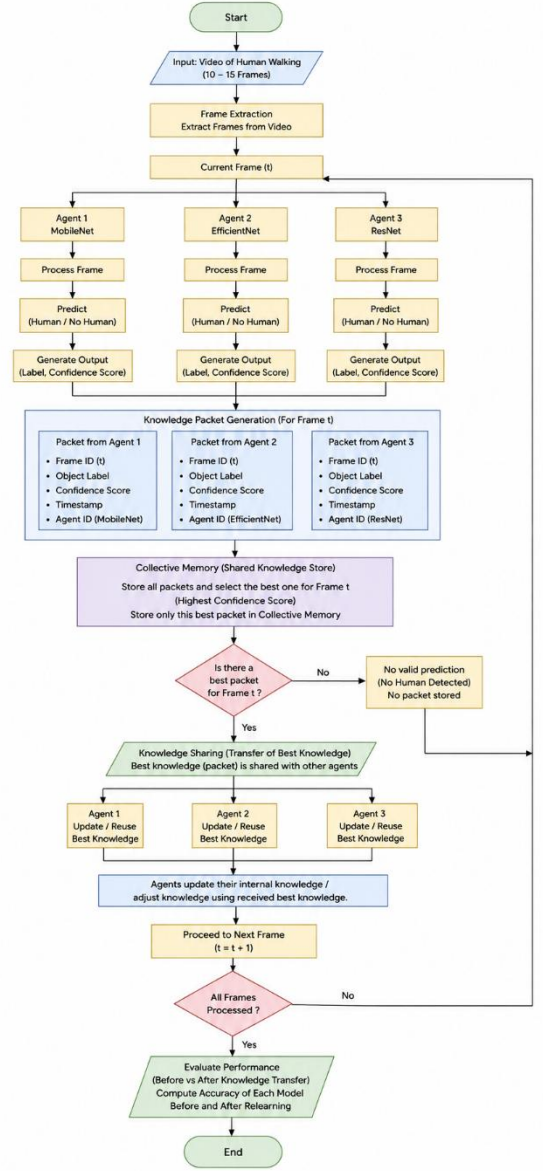


Figure 3.2: The mathematical representation of our proposed confidence based knowledge transfer framework, demonstrating knowledge generation, collective memory selection, knowledge sharing, and cross-agent knowledge reuse among heterogeneous machine learning models.

3.3 Flowchart of Proposed Method

The process starts with a video with a human strolling in it. The input video is segmented into several frames. One frame taken is used as current frame for analysis.

For each frame, all three agents individually take part in the perception process using their own machine learning model:

- Agent 1: MobileNet
- Agent 2: EfficientNet
- Agent 3: ResNet

All of the agents compare the frame and determine if there is human present in current frame. The prediction result will include the following:

- Object Label (Human / No Human)
- Confidence Score

The different architectures used by the agents can result in different confidence scores within the same frame.

Figure 3.3: Workflow of the proposed swarm-intelligent knowledge transfer framework. The process begins with frame extraction from an input video, followed by independent inference by heterogeneous agents. Knowledge packets generated by each agent are evaluated in collective memory, where the highest-confidence packet is selected and shared among participating agents. The procedure is repeated for all frames, after which performance is evaluated using accuracy and collaboration-related metrics.

Once inferred, each agent creates a knowledge packet with:

- Frame ID
- Object Label
- Confidence Score
- Timestamp
- Agent Identifier

Perception knowledge produced by each agent is represented by these knowledge packets which are used for knowledge sharing.

Any knowledge packets created are passed to the Collective Memory module. Collective memory has the role of storing collective knowledge and applying a knowledge selection policy. Confidence scores of all packets are compared for each frame. Those packets that are not remembered are those with lower scores on the confidence measure, and the packet with the highest confidence score is remembered as the Best

Knowledge Packet and remains in the collective memory. This way, the best knowledge is preserved for each frame.

After picking the best packet, it is communicated to the other agents. In this way, agents are able to access the most confident prediction in the swarm. The receiving agents update or re-use this common knowledge, which in exchange improves their perception quality, also transferring knowledge between heterogeneous machine learning models. If there's no valid prediction for a frame, no packet is stored and the system moves on to the next frame.

The above process is repeated for all extracted frames. For every frame:

- An inference is something that all agents do.
- Knowledge packets are created.
- Collective memory choose the best packet.
- Best knowledge is shared among agents.
- System advances to next frame.
- This is a "reflective" process which allows for ongoing interaction and knowledge sharing across the swarm.

Once all frames have been processed, the system makes an assessment of the performance of each model before and after the knowledge transfer. The evaluation is based on:

- Accuracy before relearning
- Accuracy after relearning
- Accuracy improvement
- Reduction in redundant inference
- Efficient knowledge storage
- Collaboration among heterogeneous models

The flowchart demonstrates how decentralized knowledge sharing enables heterogeneous machine learning models to collaboratively improve perception performance without exchanging raw video data or model parameters, thereby achieving swarm-level intelligence.

4. Results

The importance of the proposed swarm-intelligent knowledge transfer framework was evaluated using a heterogeneous multi-agent environment consisting of MobileNet-V3 Small, EfficientNet-B0, and ResNet18 models. The tests were experimented on video sequences containing human movement, where each agent independently generated predictions and confidence scores for extracted frames. The objective was to examine whether collective knowledge sharing could improve model confidence, increase knowledge reuse, and reduce redundant computational operations.



Figure 4.1 Live execution of the proposed swarm-intelligent framework showing real-time knowledge packet generation, collective memory updates, and agent collaboration.

4.1 Impact of Knowledge Transfer on Collaborative Inference

Table 1 presents the comparative performance of the framework before and after enabling knowledge transfer. Without collaborative knowledge exchange, the system operated independently and no cross-agent reuse was observed.

Table 4.1.1 Knowledge Transfer Performance Comparison

Metric	Without Knowledge Transfer	With Knowledge Transfer
Skip Rate (%)	98.75	99.17
Cross-Agent Reuse Rate (%)	0.00	64.29
Heavy Detector Calls	3	2
Calibration Success Rows	0	2

After enabling the proposed collective memory mechanism, the cross-agent reuse rate increased to 64.29%, indicating that a significant proportion of predictions were successfully supported by knowledge generated from peer agents. The skip rate increased slightly from 98.75% to 99.17%, demonstrating improved reuse of previously validated knowledge and decreasing the need for repeated processing. Furthermore, the count of heavy detector calls decreased from three to two, suggesting a reduction in computational overhead due to knowledge reuse. The calibration process also showed improvement, with successful calibration records increasing from zero to two instances after knowledge sharing was introduced. These observations indicate that the collective memory module effectively captures and redistributes useful knowledge among participating agents, enabling more efficient collaborative inference.

4.2 Improvement in Prediction Confidence

To test the influence of shared knowledge on model confidence, the same detection packet was analyzed before and after knowledge transfer. The results are given in Table 2.

Table 4.2.1 Confidence Improvement After Knowledge Sharing

Model	Before (%)	After (%)	Improvement (%)
EfficientNet-B0	50.0	84.3	34.3
MobileNet-V3 Small	50.9	85.8	34.9
ResNet18	50.0	78.0	28.0

Among the evaluated models, MobileNet-V3 Small achieved the highest improvement, with confidence increasing from 50.9% to 85.8%, corresponding to a gain of 34.9 percentage points. EfficientNet-B0 exhibited a similar improvement, increasing from 50.0% to 84.3%. ResNet18 also benefited from the shared knowledge, improving from 50.0% to 78.0%.



Figure 4.2.1 Example of a shared knowledge packet and confidence comparison before and after knowledge transfer across MobileNet-V3, EfficientNet-B0, and ResNet18 models.

The consistent improvement across the three heterogeneous architectures demonstrates that our proposed framework is not restricted to a particular model type. Instead, the transferred knowledge provides additional contextual information that assists agents in refining their predictions and increasing confidence levels.

4.3 Effectiveness of Collective Memory

The collective memory component plays a vital role in the proposed framework by storing only the highest-confidence knowledge packet for each frame. This confidence-driven selection strategy prevents the accumulation of redundant information and ensures that only the most reliable knowledge is disseminated throughout the swarm.

Experimental observations revealed that knowledge generated by one agent could be successfully reused by other agents without exchanging model parameters or training data. This characteristic distinguishes the proposed approach from conventional federated learning frameworks, which typically rely on parameter synchronization and repeated model updates.

5. Discussion

The test results show that the proposed swarm-intelligent framework can facilitate knowledge transfer between different types of ML models without retraining, parameter sharing, or sharing of raw sensor data. The knowledge packets that were generated were a semantic representation shared across all of the participating agents regardless of their underlying architecture. This collective memory mechanism proved to be very effective in choosing the most reliable packet (according to the confidence scores) and made it available to other agents to be reused. A significant increase in model confidence was observed after knowledge transfer. MobileNet and EfficientNet both achieved gains greater than 34%, and ResNet had significant gains, as well. The results display that the weaker and weaker agents may be helped by knowledge produced by the stronger agents in the swarm. The cross-agent reuse rate rose from 0% to around 64% and this demonstrates that collaboration is taking place between heterogeneous models. Meanwhile, the number of heavy inferences that run decreased, indicating fewer computations and more efficient use of resources. The experimental results corroborate the main assumption of the suggested

In collective intelligence, knowledge can be distributed in a decentralized manner in the form of semantic knowledge instead of data or model parameters. The framework allows for the effective cooperation of various agents while maintaining their autonomy, scalability, and privacy.

The findings overall validate the ability of the proposed swarm-intelligent knowledge transfer solution for better perception results, less redundant computing, more effective collaboration between heterogeneous models and a viable basis to future real-world multi-robot applications.

6. Conclusion

In this project, a swarm-intelligent approach to knowledge transfer between heterogeneous machine learning models was proposed and implemented. The framework utilizes three different deep learning models—MobileNet-V3 Small, EfficientNet-B0, and ResNet-18—as autonomous perception agents. Video frames are separately processed by each agent, and each agent independently builds up his perception knowledge, represented by object labels and confidence scores. The agents communicate with each other via structured knowledge packets rather than raw sensor data or model parameters, thus allowing model-agnostic collaboration.

A collective memory mechanism was proposed to store the best knowledge packet for each frame in terms of confidence scores. Selected knowledge is then disseminated to all agents enabling them to be able to leverage high-confidence knowledge and enhance their prediction abilities. Experimental results showed that knowledge sharing was found to substantially enhance the overall system efficiency by increasing the model confidence and decreasing the redundant inference among the heterogeneous agents, thereby improving the overall collaboration and knowledge sharing among the different agents.

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